

ConvMamba: A Data-Efficient Neural Network for Bearing Fault Diagnosis

Nguyen Thi Hue, Le Hai Anh, Vuong Viet Hung, Tran Le Duc Anh, Dao Duc Thinh, Tran Thi Thao*, Hoang Si Hong*

*School of Electrical and Electronic Engineering
Hanoi University of Science and Technology
Hanoi, Vietnam*

*Corresponding author: thao.tranthi@hust.edu.vn, hong.hoangsy@hust.edu.vn

Abstract—Bearing defects occur under harsh working conditions and due to poor maintenance. Early diagnosis of bearing failures is crucial for the safe operation of rotating machinery. Modern approaches involving the classification of bearings' vibrational signals using deep neural networks have yielded astoundingly accurate results on large benchmark datasets. However, these methods are often data-intensive, which limits their practical applications. The recent development of Hybrid-CNN-MLP was an attempt to address the challenge of data inadequacy, showing promising results. Despite this, its robustness under varying working conditions remains an issue. This paper introduces ConvMamba—a neural network architecture for the effective classification of bearing defects. We also propose an improvement in signal processing methods for more robust predictions. Compared to the previous Hybrid-CNN-MLP model, our proposed architecture offers an improvement of up to +33.56% in accuracy for tasks requiring inference under different load conditions, achieving a maximum accuracy of 99.20%.

Keywords—Bearing Fault Diagnosis, Signal Processing, Predictive Maintenance, Mamba, Deep Learning.

I. INTRODUCTION

Rolling element bearings are essential components of many rotating machines. Operating under harsh conditions and inadequate maintenance can cause defects in these components. It is crucial to predict faults in rolling elements since early diagnosis can prevent the production process from stopping abruptly, as well as protect the machine operators from potential dangers. Identifying the position of a bearing defect also aids in limiting the time and cost associated with error inspection and maintenance.

Traditionally, signal processing methods are applied to detect the presence of bearing fault with techniques such as bandpass and separation of envelope signal. Signal processing techniques can be limited since it depends heavily on the expertise of the machine operator to perform the identification. The introduction of machine learning algorithms has helped greatly automate the task of fault diagnosis. By transforming the 1-D signals to 2-D images via the short-time Fourier transform (STFT), [1] successfully performs fault identification on signals' spectrograms, using a deep convolutional neural network (CNN). Over the years, deep learning has showcased excellent accuracy on the defect classification tasks. Most notably, Zhang et al.'s [2] uses of ResNet architecture and Jiao et al.'s [3] experiments with deep coupled dense convolutional neural networks (CDCN) have presented competitive results. To overcome challenges of computational extensivity presented with deep learning, the works of [4], [5], [6], [7] aims at developing an

efficient neural network architecture that can be deployed on embedded devices, further enhance deep learning applicability in fault diagnosis.

The success of aforementioned works establishes deep learning as a promising method for detecting bearing faults in industrial applications. However, a significant drawback of deep neural networks is its reliance on a large amount of training data. Despite the existence of large benchmark datasets [8], [9], [10] on bearing vibrational signal, an algorithm needs be trained using the signal from the same bearing model it is going to be operated on to achieve high precision. Therefore, it is important that the model should effectively learned from a limited number of samples, reduce the burden on measurement and preparation of training data. Another criterion for current deep learning approaches is that the model should be well-adapted to variation in load condition. This challenge is addressed in the works of [11], [12].

As an attempt to tackle the challenge of data inadequacy, [13] introduces Hybrid-CNN-MLP – a data-efficient deep learning model for the bearing fault diagnosis. By combining ConvNet, which is a staple for fault classification task, and MLP-Mixer [14], an advancement in the field of computer vision, the approach has demonstrated some promising results: records larger than 97% accuracies when being tested on the same samples distribution as the training data. However, Hybrid-CNN-MLP has not been able to replicate similar accuracy on benchmarks that require evaluation on different load conditions. In the worst-case scenario, the model registered less than 50% accuracy, even if being trained on samples in large quantity.

Our works introduce the model of ConvMamba for classification of bearing defects. Based upon the idea of using convolutional layer for spectrogram low-level feature extraction as in Hybrid-CNN-MLP, this architecture further improves the learning capabilities by utilizing the Mamba encoder [15] block from the eponymous Vision Mamba architecture. To enhance the algorithm's robustness, an alternative approach for signal pre-processing is proposed. By making the signal segmentation technique adaptable to load conditions, while also utilizing the Constant Q-Transform for better graphic representation of measured signals, our method of signal pre-processing can produce result that is less susceptible to variation in motor speed.

Our contribution can be summarized as follows:

- We introduce the novel ConvMamba model, a fusion architecture of ConvNet and Vision Mamba for task

- of bearing fault diagnosis, provided the condition of data deficiency.
- We propose a method for data pre-processing including adaptive signal segmentation and wavelet transformation using Constant-Q Transform. Empirically, our alternative approach helps enhance robustness in model's predictions.
- We compare our results against the predecessor Hybrid-CNN-MLP on the CWRU [9] bearing dataset. For a more rigorous evaluation, the model's performance on HUST [10] bearing dataset is also going to be examined, which serves as the baseline for future works on bearing fault diagnosis on a limited amount of training data.

II. RELATED WORKS

A. Hybrid-CNN-MLP

To achieve high accuracy while only utilizing a limited amount of data, [13] proposes the Hybrid-CNN-MLP architecture: a fusion model of convolutional neural networks (CNN) and MLP-Mixer [14]. The input 1-D signal will be divided into segments before being transformed into 2-D spectrogram images using the short-time Fourier transform. The spectrogram images are then fed into a deep neural network; first, by passing through a single convolution layer to generate feature maps and reduce the data dimensionality. Next, the feature maps will be separated into patches. Each 2-D patch, excluding the channel dimensions, is flattened to a 1-D feature vector, which is used as inputs for MLP (Multi-Layer Perceptron) layers. The outputs of the final MLP layer are aggregated using a global average pooling operation to extract features for fault classification.

MLP-Mixer achieves excellent accuracy when being evaluated on the same load condition the model was trained on. However, variations in rotor speed are proved to negatively affect the model's performance. The largest accuracy discrepancy is recorded when the model is required to adapt to a lower-powered load, as evidenced by the smaller than 60% accuracy when performing inference on test samples of load 1HP after being trained on samples of load 3HP.

B. Vision Mamba

Vision Mamba is a deep neural network used for the task of image classification. The architecture is inspired by Vision Transformer (ViT) [16], starts by dividing the input image into 2-D patches. These patches are then passed through a flattening layer followed by a linear projection to be reshaped into vectors of dimension D , the *hidden dimension* of Vision Mamba model, before being added to positional

embeddings to indicate the ordinality of the patches. The vector of dimension D is augmented with a class token. After being fed through the Vision Mamba encoder, the class token is normalized and passed through an MLP (Multi-Layer Perceptron) to make the final predictions.

In Vision Mamba Block, Vision Mamba Encoder uses Bidirectional SSM. Each Vision Mamba Encoder Block consists of 2 blocks: Forward SSM and Backward SSM. This utilization of two blocks helps the model learn from information in two different directions, limiting information loss. After normalizing, the embedded patches are linearly projected into two parts, denoted as x and z , both with a dimension size of E . Then, vector x undergoes computations in both the forward and backward directions. Conv1D is applied to x to obtain for each direction. x' is then linearly projected into B_0 , C_0 and Δ_0 . Then Δ_0 is used to transform A_0 and $\bar{B}_0$. Finally, the results in $y_{forward}$ and $y_{backward}$ using the (SSM). Subsequently, gating by the $z_{forward}$ and $z_{backward}$ the outputs are combined to obtain the final token sequence.

Vision Mamba's great performance in image representation learning and domains like signal processing and audio processing makes it a strong suit for our task of fault diagnosis based on signal.

C. Constant-Q transform

Based on the success of [7] in bearing fault classification under inconsistent conditions, we apply the same method from this paper for signal - spectrogram conversion – the Constant-Q transform, using the implementation described in [17].

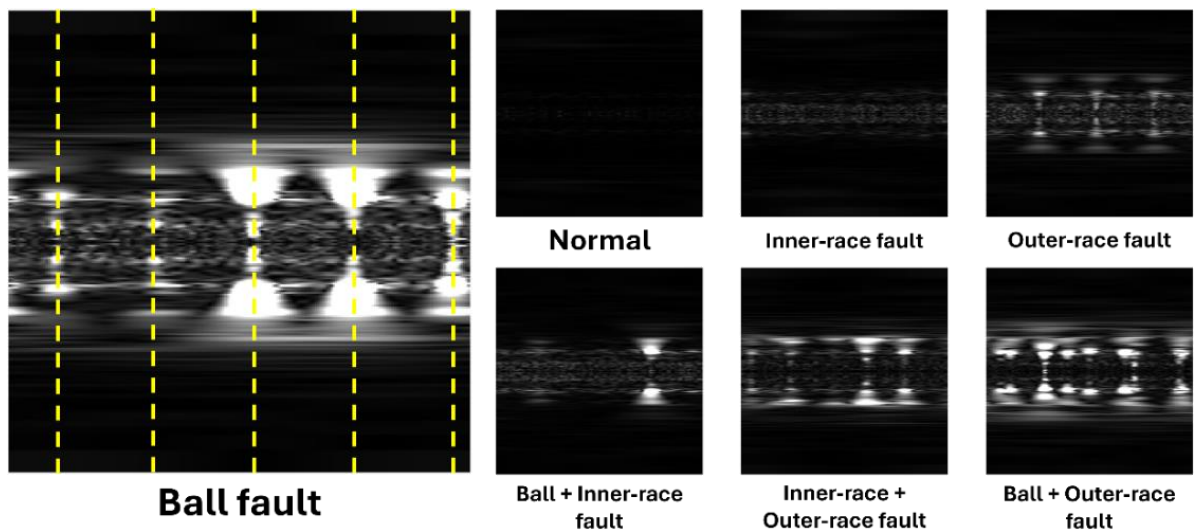


Fig. 1. Spectrograms obtained using signal measured on bearing 704 from HUST bearing dataset. The dashed yellow line highlights the impulse response from impact between rolling elements and defects

In CQT, bandwidth and sampling density are adjustable to suit the signal. CQT utilizes the short-time Fourier transform to extract frequency-related feature information from data. Below is the mathematical formulation of the CQT:

$$X^{cqt} = \frac{1}{N_k} \sum_{n=0}^{N_k-1} w_{N_k} x(n) e^{-\frac{j2\pi Q_n}{N_k}}, \quad (1)$$

where $x(n)$ is a given signal, N_k is the window function length, $w_{N_k}(n)$ is the window function, and the quality factor (Q) is defined below:

$$Q = \frac{f_k}{\delta_{f_k}}, \quad (2)$$

where f_k is the center frequency, δ_{f_k} is the bandwidth at frequency f_k .

III. PROPOSED METHODS

A. Signal Pre-processing

a) Signal Segmentation:

Before being effectively processed by the deep neural network, the input signal needs to be divided into segments and transformed to spectral images. This stage will be referred to as *signal pre-processing*. Several improvements were made to the signal pre-processing steps to ensure less performance degradation under various working conditions.

The previous work of [13] separated the signal into segments of equal length. The knowledge of bearing defect signal vibration model reveals how this approach led to the algorithm being more susceptible to the variation in motor speed. According to [18], bearing fault signal is characterized by a fault frequency f_f . The three types of defects: inner-race faults, outer-race faults, rolling-element faults have fault frequencies equivalent to BPFI (Ball Pass Frequency Inner), BPFO (Ball Pass Frequency Outer), BSF (Ball Spin Frequency), respectively. The formulas of f_f for individual type of defects are listed as follows:

$$BPFI = \frac{nf_r}{2} \left(1 + \frac{d}{D} \cos \alpha \right) \quad (3)$$

$$BPFO = \frac{nf_r}{2} \left(1 - \frac{d}{D} \cos \alpha \right) \quad (4)$$

$$BSF = \frac{Df_r}{d} \left(1 - \left(\frac{d}{D} \cos \alpha \right)^2 \right) \quad (5)$$

In which, n is the number of rolling elements, f_r is the motor rotational speed, d and D being the diameter of rolling elements and the outer race respectively. For our case of ball-typed rolling element, α is 0. From the formulation of fault frequency, for the same model of bearing, all the parameters apart from motor speed f_r are fixed. We defined T to be the inverse of f_f , and is the time interval between two consecutive impact pulses. The learning algorithm can identify the type of fault by recognizing the fault interval T from spectral image. Using the fact that has been established using (3), (4) and (5), it can be concluded that T is inversely proportional to f_r , and thus can be represented as λ/f_r , λ being a coefficient that characterized the type of defects. Let

f_s be the signal sampling frequency measured in number of samples per second and W be the designated width in number of pixels of the spectrogram to be fed to the deep neural network, a formula for how fault interval T can be represented as image pixels is derived as:

$$T_p = f_s T \frac{W}{t_p} = f_s \frac{\lambda}{f_r} \frac{W}{t_p} \quad (6)$$

In which, T_p is the fault interval measured in number of pixels on a spectral image, t_p is the number of samples in a signal segment, $\frac{W}{t_p}$ can be interpreted as the resize factor from the original spectrogram to the image of size required by the network architecture. (6) shows that T_p from the same class of defect (has the same value of λ) is dependent on f_r , which makes the learning algorithm difficult to recognize T_p since its value can be varied under changes of motor speed f_r .

Our proposed solution is to select the number of samples per segment t_p to be inversely proportional to f_r . By substituting t_p as ψ/f_r to (6), we obtain:

$$T_p = f_s \frac{\lambda}{f_r} \frac{f_r}{\psi} W = f_s W \frac{\lambda}{\psi} \quad (7)$$

Eq.(7) is removed of its dependent on f_s . This form of T_p is therefore consistent under variation in load conditions. For our experiment, ψ is selected to be equal to sampling frequency f_s , thus makes the formula for t_p and T_p be, respectively:

$$t_p = \frac{f_s}{f_r} \quad (8)$$

$$T_p = \lambda W \quad (9)$$

Eq.(8) can be interpreted as t_p , the number of samples in a rotational period. The selection of ψ parameters is informed by the fact that a rotational period capture all frequency characteristics that describe the bearing fault signal using the model of [19]. The technique of adaptive signal segmentation helps the model achieve higher accuracy than the predecessor, as described in Table I.

b) Spectrogram Transform:

After being applied adaptive segmentation, individual segments are being made into spectrograms using Constant-Q transform as described in Section II-C, the resulting spectrogram associates each pixel with a complex value. Since the fault signal characteristics are invariant with respect to phase, it is natural to take the absolute value of the spectrogram, leaving it to be a 2-D array of real value. [13] uses STFT with 75% window length overlap, which results in spectral image with temporal resolution of just 4 pixels. When being resized to a 32×32 image, most of the pixels in the time axis are interpolated, thus conveying limited information. Our proposed method of spectrogram transformation results in an image with resolution that is adaptable to sampling frequency of the measured signal. For example, in HUST bearing dataset, spectrograms retain a resolution of 122×118 pre-resize. Fig. 1 demonstrates our approach ability to retain signal features more effectively.

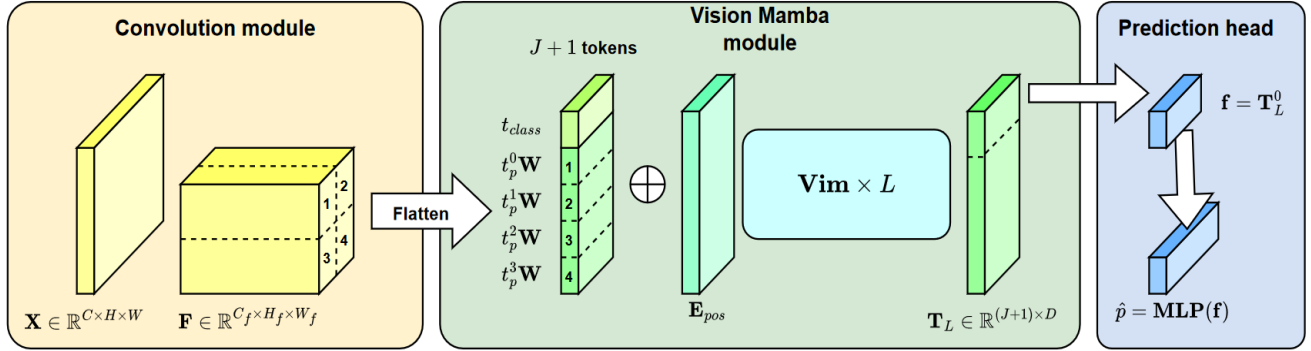


Fig. 2. The overview of the proposed ConvMamba deep neural network architecture

B. ConvMamba

To classify spectrograms from signal pre-processing stage, we introduce the ConvMamba model (Fig. 2) – a fusion architecture of the popular ConvNet and Vision Mamba [15]. The intuition behind this choice of architecture is that the model needs different methods of handling time-domain and frequency-domain characteristics. A more detailed description and motivation behind the architectural design will be explained in this section.

a) Convolution Module:

Fault characteristics of signals should be invariant with respect to shift in phase. In image processing, this property is also referred to as *translational equivariance*. Since convolution operation consists of applying a same set of weight across images using the sliding window approach, it is translational equivariance, thus, is a strong suit for extracting time-domain features. The input image $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ is first passed through convolution layer(s) and transformed into a feature map $\mathbf{F} \in \mathbb{R}^{C_f \times H_f \times W_f}$. Note that pooling can be applied in this stage so that the dimensionality of the feature map may differ from that of the input image.

b) Vision Mamba (Vim) Module:

As explained, CNN effectiveness in extraction of time-domain features is the reason for the architecture dominance in the fault diagnosis based on spectrograms. However, spectrograms' property of translational equivariance does not well-translate into frequency domain, since a translation in time domain would equate a phase rotation in frequency domain due to Fourier transform *time shifting* property. As a result, applying convolution operation would introduce *inductive bias* in the context of frequency-domain characteristic. To address the problem of inductive bias in CNN, numerous architectures, such as [14], [15], [16], have been proposed. To further process the feature maps from the previous convolution stage, an architecture largely based on Vision Mamba [15] is selected for our method. The feature maps $\mathbf{F} \in \mathbb{R}^{(C_f \times H_f \times W_f)}$ will be splitted along spatial dimensions into J patches of size $C_f \times P \times P$. All the image patches will be flattened and concatenated into a sequence of patches $\mathbf{x}_p \in \mathbb{R}^{J \times (C_f \times P^2)}$. Each patch t_p^j will be linearly projected via matrix multiplication with a shared weight matrix $\mathbf{W} \in \mathbb{R}^{(C_f \times P^2) \times D}$. To help the model adaptive to the number of patches, a class token $t_{class} \in \mathbb{R}^D$ is concatenated to

the sequence of patches. Positional encoding $\mathbf{E}_{pos} \in \mathbb{R}^{(J+1) \times D}$ similar to that in [16] is added to the patch sequence, the result is the sequence T_0 expressed using the formula:

$$\mathbf{T}_0 = [t_{class}; t_p^0 \mathbf{W}; t_p^1 \mathbf{W}; t_p^2 \mathbf{W}; \dots; t_p^{J-1} \mathbf{W}] + \mathbf{E}_{pos} \quad (10)$$

The sequence T_{l-1} will be used as input for the l^{th} Mamba encoder (Vim) [15] block with residual connection [20] to obtain the sequence T_l , as described in Section II-B. This process is replicated L times for the overall architecture:

$$\mathbf{T}_l = \mathbf{Vim}(\mathbf{T}_{l-1}) + \mathbf{T}_{l-1} \quad (11)$$

c) Prediction Head:

In the final sequence T_L , the normalized class token be used as feature to extract class prediction for the type of fault:

$$\mathbf{f} = \mathbf{Norm}(\mathbf{T}_L^0) \quad (12)$$

$$\hat{p} = \mathbf{MLP}(\mathbf{f}) \quad (13)$$

In inference mode, prediction will be obtained as the element of highest value in the prediction vector $\hat{p}$:

$$\hat{y} = \mathbf{argmax}(\hat{p}) \quad (14)$$

While in training mode, $\hat{p}$ will be used as logits for the loss function to be optimized as described in Section IV.

IV. IMPLEMENTATION DETAILS

a) Detailed Architecture:

The architecture of the ConvMamba model used for our experiment is detailed as follows:

- Spectral images $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ are resized to the shape of 32×32 .
- $\mathbf{X}$ are then passed through a convolution layer with the kernel size of 3×3 to create feature maps $\mathbf{F} \in \mathbb{R}^{C_f \times H_f \times W_f}$ with channel dimension of 64. Batch normalization and ReLU activation is applied to feature maps, before max pooling with kernel size of 2×2 is performed to reduce the feature maps' size to $64 \times 16 \times 16$.

- The outputs from the pooling layers are divided into 4 patches, each of size $64 \times 4 \times 4$ to become input for $L = 8$ Mamba encoder (Vim) blocks. The Mamba encoders' hidden dimension D is selected to be 32.

b) Model Training:

These architecture hyperparameters are found empirically by monitoring the validation accuracy after training the model for one epoch.

The ConvMamba model is trained by optimizing the cross-entropy loss:

$$l(\hat{p}, y) = -\frac{1}{N} \sum_{n=1}^N \frac{\exp(\hat{p}_{n,y_n})}{\sum_{c=1}^C \exp(\hat{p}_{n,c})} \quad (15)$$

In which, N is the number of samples, C is the number of classes, the loss l is a function of model's outputs $x \in \mathbb{R}^{N \times C}$ and ground-truth label $y \in \mathbb{R}^N$.

We use the Adam optimizer with the initial learning rate lr_0 of $1e-3$. In addition, the exponential learning rate scheduler is also applied to help the model converges better in later training epochs:

$$lr_t = lr_0 \times \gamma^t, \quad (16)$$

where lr_t is the learning rate at epoch t , γ is set to 0.9 for all experiments.

Training is conducted on the GPU NVIDIA GeForce GTX 1650 with Max-Q Design with a VRAM of 4GB.

V. EXPERIMENTS AND RESULTS

A. Experiment Setup

The proposed model will be trained on subsets from each of the datasets [8], [9], [10]. The training set will only contain data samples corresponding to one load condition. Similar to [13], model trained on different number of samples will be evaluated independently to examine its ability of learning from a limited amount of data. More specifically, for each dataset, per one load condition, ConvMamba will be trained on subsets of 3000, 300 and 90 samples. To circumvent the problem of data imbalance, most prominent in case of samples scarcity, the balance sampling strategy is applied to guarantee each target class contain a roughly equal amount of samples. It is observed that subsets of fewer samples requires higher number of training epochs for the model to generalize. Thus in cases of 3000, 300 and 90 samples, the model is trained for 100, 50 and 30 epochs, respectively. The specific value for number of epochs are found empirically.

After the training phase, the proposed model is evaluated on all load conditions available from the dataset. A fixed number of 500 testing samples are selected per load condition for accurate evaluation. We acknowledge the signal segmentation strategy in [13] can lead to data leakage, since training and testing segments can overlap. Accordingly, signal will be divided into non-overlapping segment like [6] for a more representative result.

B. Evaluation Metrics

To determine the proposed model's performance, we use the accuracy metric which is defined as:

$$Accuracy = \frac{\sum_{n=1}^N 1\{\hat{y} = y\}}{N}, \quad (17)$$

where N is the total amount of samples and $1\{\hat{y} = y\}$ is the identity function, evaluated to 1 had the predicted label match the ground-truth one ($\hat{y} = y$) and to 0 otherwise.

C. Comparative results on CWRU bearing dataset

To highlight the advantages our proposed method have over the predecessor [13], we compare these models performance on the Case Western Reserve University (CWRU) bearing dataset. For defect bearing cases, signals of sampling rate 12kHz with fault occurring on the drive-end of the motor are selected. Only the records from the drive-end sensor will be used for experiments. CWRU dataset contains baseline signal of motor working under normal conditions as well as bearing with defects in one of 3 types: inner-race (IR), outer-race (OR) and balls (B). For each of the fault type, defects come at 3 fault diameters: 0.007 inches, 0.014 inches and 0.021 inches. The goal of the algorithm is to correctly classify the signal sample into one of the 10 classes based on their fault type and fault size as in Table I.

TABLE I. GROUND-TRUTH CLASSES FROM CWRU BEARING DATASET

	Normal	B	IR	OR
0	0	-	-	-
0.007	-	1	4	7
0.014	-	2	5	8
0.021	-	3	6	9

From the training process, 9 models are obtained with respect to each training load and number of samples. Each trained model will be evaluated on 3 different load conditions. The experiment results are shown in Table II, in which the method that achieved testing accuracy, either Hybrid-CNN-MLP or our model, is highlighted. Based on Table II, it is discernible that our architecture ConvMamba demonstrates better performance, record higher accuracy in 18 out of 27 categories. The most significant metric improvement are seen in inference on 1HP load with model trained on 3HP load, in which a different of up to +27% shown compare to the previous method, thus demonstrate our model ability of overcome the most prominent deficit in current diagnosis methods. Higher accuracy is also exhibited with model trained on 2HP load, although the discrepancy is not as significant as the previously shown cases. Hybrid-CNN-MLP demonstrates better results on tasks using model trained on load 1HP, but it is worth noticing that the advantage the previous approach had over ours is not as evident, as ConvMamba still achieve higher than 70% accuracy at experiments. signal will be divided into non-overlapping segment similar to [6] for a more representative result.

TABLE II. MODELS ACCURACY (%) ON CWRU DATASET. A COMPARISON BETWEEN HYBRID-CNN-MLP[13] AND CONVMAMBA (OURS) METHOD

Dataset		Number of training samples					
		90		300		3000	
Train load (HP)	Test load (HP)	Hybrid-CNN-MLP	Ours	Hybrid-CNN-MLP	Ours	Hybrid-CNN-MLP	Ours
1	1	98.12	91.04	98.47	98.29	99.76	100.00
	2	93.20	84.56	91.84	94.86	97.68	96.91
	3	77.28	73.36	82.00	78.80	94.90	78.48
2	1	82.60	89.23	85.14	95.80	77.64	96.57
	2	97.89	95.54	98.69	99.68	99.85	99.89
	3	87.92	90.16	88.90	97.90	94.24	99.20
3	1	56.96	78.54	50.12	77.54	45.44	79.01
	2	89.56	91.18	81.88	88.56	96.20	91.99
	3	97.03	98.24	98.45	100.00	99.36	100.00

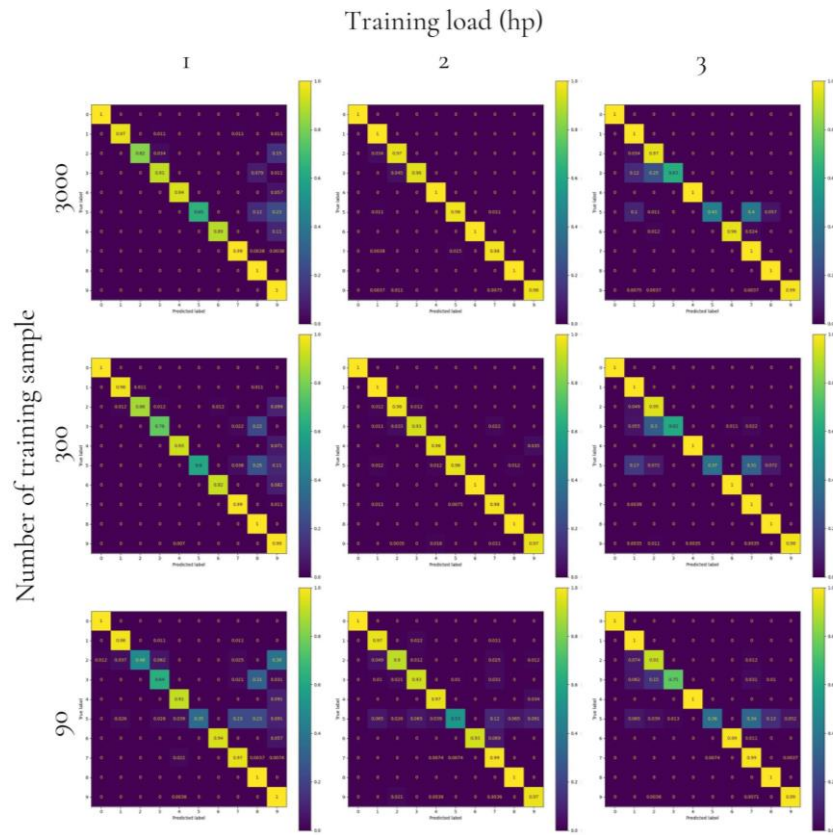


Fig. 3. Confusion matrices for CWRU dataset experiments by proposed method

In this research, the ConvMamba model's diagnostic performance on the CWRU dataset is scrutinized via a collection of confusion matrices (Fig. 3), articulating classification accuracy across varying sample sizes (90, 300, 3000) and distinct load conditions (1HP, 2HP, 3HP). These matrices, expressed as percentages, serve to illustrate the precision of the model's predictions, and offer insight into the classification interplay between different fault types.

For the smallest sample set (90), under a load of 1HP, the model manifests high accuracy, particularly for non-adjacent classes, with most true positive rates soaring above the 90% mark. Notably, however, there's a tangible confusion between adjacent classes, such as 3 and 4, with a misclassification rate in the vicinity of 12-15%, and between 7 and 8, with errors surging to approximately 25%.

Elevating the sample count to 300 under a 2HP load condition, the model maintains a robust diagonal presence in the confusion matrix, indicating sustained high accuracy. Yet, it encounters persistent misclassification between certain classes, such as 5 and 6, albeit reduced to around 12%. The confusion between classes 7 and 8 remains more pronounced, suggesting a potential insensitivity of the model to subtle variations between these classes under medium load. When the sample size reaches its apex (3000), under the heaviest load of 3HP, the ConvMamba model reaches its stride, delivering near-perfect classification for most classes. Despite the model's high fidelity, there is still a smattering of confusion in closely related fault types, with misclassification rates dwindling to approximately 7% for classes such as 5 and 6. These minimal errors point to the

model's substantial capacity for learning and generalizing from extensive data, even in the face of increased operational stressors.

D. Baseline results on HUST bearing dataset

To further examine the performance of our proposed model. We also evaluate the ConvMamba model on the HUST bearing dataset [10]. HUST bearing contains signal samples belonging to one of the 7 classes: normal (N), ball (B) fault, inner-race (I) fault, outer-race (O) fault, as well as combination of the type of defects (IB, IO and OB). To the best of our knowledge, ConvMamba is the first model to successfully tackle the task of classifying signal containing potential combined faults on the HUST bearing dataset. Our results (Table III) will serve as the baseline for future studies related to diagnosis of both single and double faults on HUST bearing dataset. The dataset contains signal measured from 5 different models of bearing with inner-race diameter of: 20mm (type 4), 25mm (type 5), 30mm (type 6), 35mm (type 7), 40mm (type 8); recorded at 3 different load conditions of 0HP, 2HP and 4HP. Since there are missing samples for bearing of type 4 and 8, training samples will only be obtained from the 3 remaining bearing models. The results for our experiments are detailed in Table III.

TABLE III. GROUND-TRUTH CLASSES FROM HUST BEARING DATASET

Dataset		Number of training samples		
Train load (HP)	Test load (HP)	90	300	3000
0	0	86.13	97.36	99.90
	2	72.98	82.71	90.60
	4	70.66	81.14	89.95
2	0	72.71	85.43	90.19
	2	83.35	95.73	99.18
	4	78.45	91.26	95.76
4	0	67.06	80.66	83.59
	2	74.33	86.52	92.10
	4	84.26	96.34	99.66

Across different tasks, ConvMamba records promising accuracy of larger than 67%. If only evaluation on the same load condition is taken into account, our method consistently achieved accuracy of larger than 83%. Variation in test load may cause degradation in performance but generally, the discrepancy is smaller than 16%. A more significant difference in load condition results in less satisfactory results, as evident by the fact that model trained on 0HP and 4HP load has an easier time adapting to test load of 2HP.

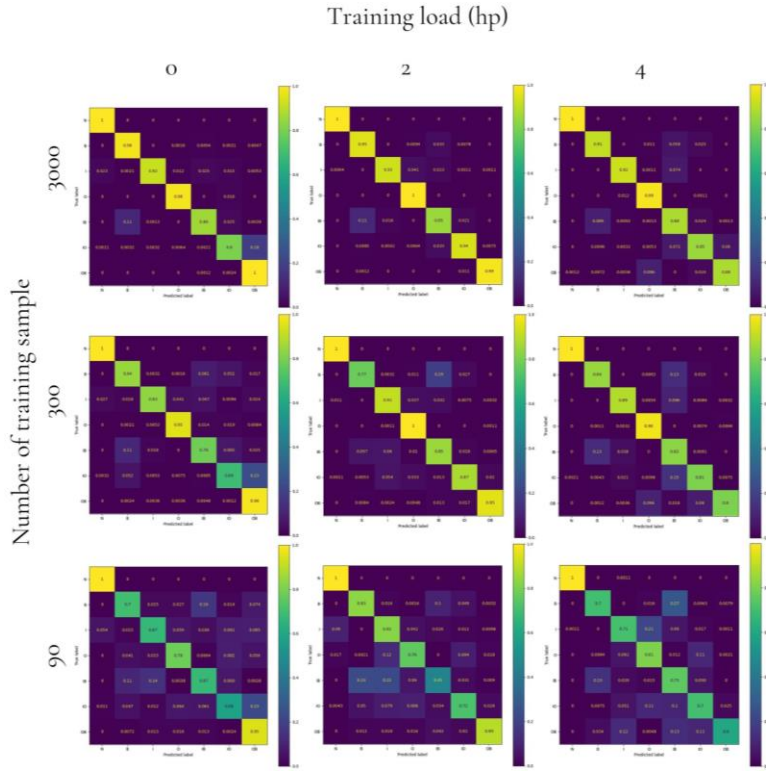


Fig. 4. Confusion matrices for HUST bearing dataset experiments by proposed method

Through rigorous evaluation, the confusion matrices (Fig. 4) present a nuanced picture of the model's adeptness at fault classification. At the outset, with the lowest sample volume of 90, the model's prowess is evident as it achieves a remarkable feat of perfect classification (100%) for the normal (N) state bearings. However, the subtleties of bearing defects present a greater challenge, as shown by a confusion rate of up to 27% between Ball (B) and Inner Race (I) defects. Scaling up to the 300-sample spectrum, the ConvMamba model sustains commendable accuracy, with a notable true positive rate of 91% for Inner Race (I) defects.

E. Ablation Studies

Considerate hyperparameter selection is therefore necessary to ensure high performance. To search for optimal model number of encoders, the ConvMamba model is trained on 3000 spectrograms of 1HP load from the CWRU dataset for 100 epochs. It is then evaluated under 3 different loads condition, the setting that yielded the highest average accuracy will be selected for benchmark results.

TABLE IV. ABLATION STUDIES ON NUMBER OF MAMBA ENCODERS

Number of encoders	Test load 1	Test load 2	Test load 3	Average
1	81.55	97.51	100.00	93.02
2	79.88	97.53	100.00	92.47
3	80.52	97.49	100.00	92.67
4	85.93	98.69	100.00	94.87
5	83.94	98.50	99.67	94.04

Table IV shown a surge predictive performance as the number of decoders increases from 1 to 4. The dip in accuracy in the case of 5 decoders suggests that under this setting, over-parametrization starts to occur. Thus, 4 is chosen as the optimal model depth.

F. Model Complexity

We perform comparison on number of parameters on of our proposed ConvMamba and its predecessor Hybrid-CNN-MLP method. Table V shows parameters comparison breakdown by sub-modules.

TABLE V. COMPARISON ON NUMBER OF PARAMETERS

Hybrid-CNN-MLP	Number of Parameters	ConvMamba	Number of parameters
Conv	640	Conv	640
Patch Embedding	1,048,832	Patch Embedding	2570
MLP-Mixer (×8)	67092×8	Mamba Encoder (×4)	(9984×4)
Prediction Head	2570	Prediction Head	330
Overall	1,588,778	Overall	43,476

MLP-Mixer and Mamba Encoder block is the main architectural differences between two models. It is evident that our ConvMamba encoder utilizes fewer parameters by having smaller embedding size (64 versus 128) and fewer number of depth layer (4 versus 8), thus demonstrates the model learning efficiency.

VI. CONCLUSION

In this research, a method for data-efficient detection of bearing defects based on motor vibrational signal is introduced. The proposed ConvMamba model can effectively learn both signal time-domain and frequency-domain characteristics leveraged by signal pre-processing techniques for a more robust prediction. Our method achieves higher accuracy than its predecessor, especially on tasks that required evaluation on various load conditions. Despite significant improvements, the accuracy of less than 80% on several benchmarks means that there is room for improvement in future research.

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